QUESTIONS AND SOLUTIONS

SECTION A : 5 marks each

1. Find the value of $F(13.333)$ for the function F defined by $F(x) = 2x$ for $x < 1$ and $F(x) = F(x-1)$ for $x \geq 1$.

Solution

$$\begin{aligned} F(13.333) &= F(12.333) \\ &= \ldots \\ &= F(1.333) \\ &= F(0.333) \\ &= 2 \cdot 0.333 \\ &= 0.666 \end{aligned}$$

2. Let x, y, z be positive real numbers. Use the **AM-GM** (arithmetic mean - geometric mean) inequality to prove:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{2}{x+y} + \frac{2}{y+z} + \frac{2}{x+z}.$$

Solution

From the **AM-GM** (arithmetic mean - geometric mean) inequality, we have:

$$(x+y)^2 \geq 4xy,$$

and from this inequality one gets:

$$\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}. \tag{1.1}$$

Similarly,

$$\frac{1}{y} + \frac{1}{z} \geq \frac{4}{y+z}, \quad \text{and} \quad \frac{1}{x} + \frac{1}{z} \geq \frac{4}{x+z}. \tag{1.2}$$

Now, summing up all the inequalities in (1.1) and (1.2), together with some simplification, yields:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{2}{x+y} + \frac{2}{y+z} + \frac{2}{x+z},$$

as required.

3. Prove that for any integer n, the number

$$2023^{5^{n+1}} + 2023^{5^n} + 1$$

is not a prime number.

Solution

Let $x = 2023^{5^m}$. Then

$$2023^{5^{n+1}} + 2023^{5^n} + 1 = x^5 + x + 1 = (x^2 + x + 1)(x^3 - x^2 + 1),$$

and since both factors are clearly greater than 1, the conclusion that $2023^{5^{n+1}} + 2023^{5^n} + 1$, is not a prime number follows.

Alternatively, Using the fact that $2023 \equiv 1(\mod 3)$, we have that $2023^{5^n} \equiv 1^{5^n} \equiv 1(\mod 3)$, similarly, $2023^{5^{n+1}} \equiv 1^{5^{n+1}} \equiv 1(\mod 3)$,
So, $2023^{5^{n+1}} + 2023^{5^n} + 1 \equiv 1 + 1 + 1 \equiv 3 \equiv 0(\mod 3)$
Therefore $2023^{5^{n+1}} + 2023^{5^n} + 1$ is divisible by 3 for any n, Hence, it is not a prime.

4. Triangle ABC has side length equal to a, b, c. Find a necessary and sufficient condition on the angles A, B, C of the triangle ABC such that $a^2,\ b^2,\ c^2$ can be side lengths of another triangle.

Solution

The positive real numbers a^2, b^2, c^2 can be the side length of a triangle if and only if:

$$a^2 + b^2 > c^2, \quad b^2 + c^2 > a^2, \quad \text{and} \quad a^2 + c^2 > b^2. \tag{1.3}$$

By the Cosine Rule:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \quad \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \quad \text{and} \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab},$$

we get that (1.3) is equivalent to:

$$\cos A > 0, \quad \cos B > 0, \quad \text{and} \quad \cos C > 0.$$

Thus, the necessary and sufficient condition is that triangle ABC is acute, that is, all the angles of triangle ABC are less than 90°.

5. Find the product of the roots of the equation:

$$y^2 - 31y + 220 = 2^y(31 - 2y - 2^y).$$

Solution

The equation $y^2 - 31y + 220 = 2^y(31 - 2y - 2^y)$ is equivalent to the equation:

$$(y + 2^y)^2 - 31(y + 2^y) + 220 = 0. \tag{1.4}$$

Consequently, the roots of (1.4) satisfy:

$$y + 2^y = 11 \quad \text{or} \quad y + 2^y = 20. \tag{1.5}$$

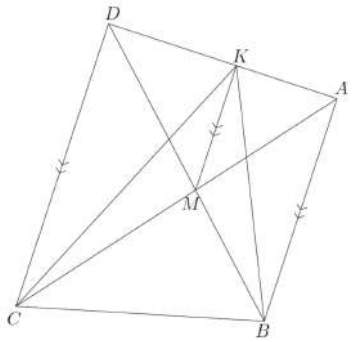
Since the function $f(y) = y + 2^y$, for $y \in \mathbb{R}$ is increasing, then each of the equations in (1.5) has at most one root. One can easily verify that $y_1 = 3$ satisfies the first equation in (1.5) and $y_2 = 4$ satisfies the second equation in (1.5). Thus, the product of roots is given by:

$$y_1 y_2 = 3 \times 4 = 12.$$

6. Let $ABCD$ be a quadrilateral such that $\angle BAD = \angle ADC = 90^o$. Diagonals AC and BD meet at M. Let K be a point on side AD such that $\angle ABK = \angle DCK$. Prove that KM bisects $\angle BKC$.

Solution

Since $\angle BAD = \angle ADC = 90^o$ and $\angle ABK = \angle DCK$, triangles CDK and BAK are similar, so we have $CD/AB = DK/KA$. Since CD and AB are parallel, triangles CDM and ABM are also similar, we also have $CD/AB = DM/MB$. Hence, $DK/KA = DM/MB$. Therefore, CD, AB, and KM are all parallel. Finally, $\angle MKC = \angle DCK = \angle ABK = \angle MKB$, as required.



7. An integer has 3 digits in base 10. When it is interpreted in base 6 and multiplied by 4, the result is 2 more than 3 times the same number when it is interpreted in base 7. What is the largest integer, in base 10, that has this property?

Solution

Let the three-digit number be rst. Since $4 \times rst_6 = 3 \times rst_7 + 2$, we have:

$$\begin{aligned} 4(36r + 6s + t) &= 3(49r + 7s + t) + 2 \\ 144r + 24s + 4t &= 147r + 21s + 3t + 2 \\ \Rightarrow \quad 3s + t &= 3r + 2. \end{aligned}$$

Hence, 3 divides $t - 2$. That is, $t - 2 = 3k$ for $k = 0, 1, 2$. Since $t < 6$, we have that $t - 2 = 0$ and $t - 2 = 3$ or $t = 2$ and $t = 5$. If $t = 2$, then $3t = 3s$ or $r = s$. If $t = 5$, then $3s + 3 = 3r$ or $r = s + 1$. Since all digits are at most 5, the largest required number, in base 10, is $rst = 552$.

8. If a, b, c are real numbers with $a - b = 4$, find the maximum value of $ac + bc - c^2 - ab$.

Solution

$$\begin{aligned} ac + bc - c^2 - ab &= (a - c)(c - b) \\ &= \left[\sqrt{(a - c)(c - b)}\right]^2 \\ &\leq \left[\frac{(a - c) + (c - b)}{2}\right]^2 \\ &= \left[\frac{a - b}{2}\right]^2 \end{aligned}$$

$$= \left[\frac{4}{2}\right]^2$$
$$= 4$$

The maximum is attained when $a = 4, b = 0$, and $c = 2$.

9. Malcolm writes a positive integer on a piece of paper. Malcolm doubles this integer and subtracts 1, writes this second integer on the same piece of paper. Malcolm then doubles the second integer and adds 1, writes this third integer on the paper. If all of the numbers Malcolm writes down are prime, determine all possible values for the first integer.

Solution

Let n be the first integer. Then $2n - 1$ is the second and $4n - 1$ is the third. We can write n as $3k, 3k + 1$, or $3k + 2$ for some non-negative integer k.

When $n = 3k$, we must have $k = 1$, since n is a prime. In this instance the three number are $3, 5, 11$, which are all prime.

When $n = 3k + 1$, we have $4n - 1 = 12k + 3 = 3(4k + 3)$. Since this is a multiple of 3, it must be 3, and we have $k = 0$. This means $n = 1$, and the first number written down is not prime.

When $n = 3k + 2$, we have $2n - 1 = 6k + 3 = 3(2k + 1)$. Since this is a multiple of 3, it must be 3, and we have $k = 0$. In this instance, the three numbers are $2, 3, 7$, which are all prime.

Thus, the first integer is either 2 or 3.

10. Consider a function $f : (0, \infty) \to \mathbb{R}$ and a positive real number $a > 0$ such that $f(a) = 1$. Prove that if

$$f(x)f(y) + f\left(\frac{a}{x}\right) f\left(\frac{a}{y}\right) = 2f(xy)$$

for all $x, y \in (0, \infty)$, then f is a constant function.

Solution

Setting $x = y = 1$ yields:

$$f^2(1) + f^2(a) = 2f(1), \quad \text{or} \quad (f(1) - 1)^2 = 0, \quad \text{so} \quad f(1) = 1.$$

Substituting $y = 1$, gives:

$$f(x)f(1) + f\left(\frac{a}{x}\right) f(a) = 2f(x), \quad \text{or} \quad f(x) = f\left(\frac{a}{x}\right), \quad x > 0.$$

Take now, $y = \frac{a}{x}$ and observe that:

$$f(x)f\left(\frac{a}{x}\right) + f\left(\frac{a}{x}\right) f(x) = 2f(a), \quad \text{or} \quad f(x)f\left(\frac{a}{x}\right) = 1.$$

Therefore, $f^2(x) = 1$, $x > 0$.

Now, set $x = y = \sqrt{t}$, which gives:

$$f^2(\sqrt{t}) + f^2\left(\frac{a}{\sqrt{t}}\right) = 2f(t), \tag{1.6}$$

and because the left-hand side of (1.6) is positive, it follows that f is positive and $f(x) = 1$ for all x. Then, f is a constant function, as required.

Section B: 10 marks each

11. If a cock costs \$ 5, a hen costs \$ 3, and three chicks together cost \$ 1, how many cocks, hens, and chicks, totalling 100, can be bought for \$ 100?

Solution

Let x be the number of cocks, y be the number of hens, and z be the number of chicks bought. This problem can be then written as:

$$5x + 3y + \frac{1}{3}z = 100 \quad \text{and} \quad x + y + z = 100.$$

Eliminating one of the unknowns, we are left with a linear Diophantine equation in the other two variables. In particular, since $z = 100 - x - y$, we have $5x + 3y + \frac{1}{3}(100 - x - y) = 100$, or

$$7x + 4y = 100. \tag{1.7}$$

The equation (1.7) is a linear Diophantine equation with the general solution:

$$x = 4t, \quad y = 25 - 7t, \quad \text{so that} \quad z = 75 + 3t,$$

where t is an arbitrary integer. For this problem, t must be chosen to satisfy simultaneously the inequalities:

$$4t > 0 \quad 25 - 7t > 0, \quad \text{and} \quad 75 + 3t > 0. \tag{1.8}$$

The last two inequalities in (1.8) are equivalent to: $-25 < t < 3\frac{4}{7}$. From the first inequality in (1.8), we have that $t > 0$, then the possible integer values of t are: $t = 1, 2, 3$; leading to the solutions:

$$\begin{aligned} t = 1: &\quad x = 4,\ y = 18,\ z = 78; \\ t = 2: &\quad x = 8,\ y = 11,\ z = 81; \\ t = 3: &\quad x = 12,\ y = 4,\ z = 84. \end{aligned}$$

12. If the numbers a_n are defined inductively by $a_1 = 11$, $a_2 = 21$, and $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \geq 3$, prove that $a_n = 5(2^n) + 1$ holds for all $n \geq 1$.

Solution

We prove the result by induction on n. For $n = 1$ and $n = 2$, we have:

$$a_1 = 11 = 5(2^1) + 1 = 10 + 1 = 11 \quad \text{and} \quad a_2 = 21 = 5(2^2) + 1 = 20 + 1 = 21.$$

Hence the formula holds in both cases, and this provides the basis for this induction.

For the induction step, we choose an integer $k \geq 3$ and assume that the formula is true for $n = 1, 2, \ldots, k - 1$. Then, in particular:

$$a_{k-1} = 5(2^{k-1}) + 1 \quad \text{and} \quad a_{k-2} = 5(2^{k-2}) + 1.$$

By the way these numbers are defined, we have:

$$\begin{aligned} a_k = 3a_{k-1} - 2a_{k-2} &= 3\left(5(2^{k-1}) + 1\right) - 2\left(5(2^{k-2}) + 1\right) \\ &= 15(2^{k-1}) + 3 - 5(2^{k-1}) - 2 = 10(2^{k-1}) + 1 = 5(2^k) + 1. \end{aligned}$$

Thus, $a_k = 5(2^k) + 1$ and since the formula is true for $n = k$ whenever it is true for $n = 1, 2, \ldots, k - 1$; we conclude by strong induction that $a_n = 5(2^n) + 1$ holds for all $n \geq 1$.

Alternatively, From the recurrence relation $a_n = 3a_{n-1} - 2a_{n-2}$, we get the the characteristic polynomial $x^2 - 3x + 2$, whose roots are $x = 1$ and $x = 2$.

Using the general theory of recurrence relations, we have the general (Binet's formula):

$$a_n \quad = \quad a \cdot (1)^n + b \cdot (2)^2 \tag{1.9}$$

for some contestants a and b that are determined using the initial conditions:

For $n = 1$, we see that $a_1 = 11$ and (1.9) becomes:

$$11 \quad = \quad a + 2b \tag{1.10}$$

For $n = 2$, we see that $a_2 = 21$ and (1.9) becomes:

$$21 \quad = \quad a + 4b \tag{1.11}$$

Solving (1.10) and (1.11) simultaneously gives:

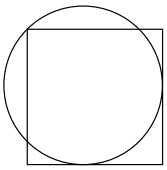
$(1.11) - (1.10) \Rightarrow 2b = 10 \Rightarrow b = 5$

put in (1.9) $\Rightarrow 11 = a + 2 \cdot 5 \Rightarrow a = 11 - 10 = 1$

So substituting back in (1.9), we get:

$$a_n = 5(2)^n + 1 \quad \text{for all} \quad n \geq 1$$

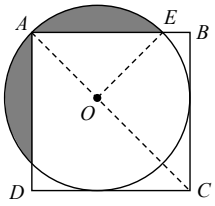
13. Pugsley tries to fit a round circle into a square hole, but part of the circle sticks out. What is the total area of the two excess regions that Pugsley will carve from the circle? The circle has radius 10, and Pugsley has it placed so that it passes through one corner of the square and is tangent to two sides of the square.



Solution

Let $ABCD$ be the square with A on the circle.

Let O be the center of the circle. Let AB intersect the circle at E.



Draw the diagonal AC of the square. Obviously, O is on AC.

With $\angle EAO = 45^\circ$ and $OA = OE$, AOE is an isosceles right triangle.

Then we have $\angle AOE = 90^\circ$.

The area of the shaded circular segment AE is

$$\frac{1}{4} \cdot 10^2\pi - \frac{1}{2} \cdot 10^2 = 25\pi - 50.$$

The total area of the shaded regions is

$$2 \cdot (25\pi - 50) = 50\pi - 100$$

14. Prove that if a, b, c are nonzero real numbers with $a + b + c = 0$, then

$$\frac{a^2+b^2}{a+b} + \frac{b^2+c^2}{b+c} + \frac{a^2+c^2}{a+c} = \frac{a^3}{bc} + \frac{b^3}{ac} + \frac{c^3}{ab}.$$

Solution

Since $a + b + c = 0$, we obtain:

$$a + b = -c, \quad b + c = -a, \quad \text{and} \quad a + c = -b,$$

or by squaring and rearranging we get:

$$a^2 + b^2 = c^2 - 2ab, \quad b^2 + c^2 = a^2 - 2bc, \quad \text{and} \quad a^2 + c^2 = b^2 - 2ac.$$

Thus, the given equation in question is equivalent to:

$$\frac{c^2-2ab}{-c} + \frac{a^2-2ab}{-a} + \frac{b^2-2ac}{-b} = \frac{a^3}{bc} + \frac{b^3}{ac} + \frac{c^3}{ab},$$

and consequently equivalent to:

$$-(a+b+c) + 2\left(\frac{ab}{c} + \frac{bc}{a} + \frac{ac}{b}\right) = \frac{a^3}{bc} + \frac{b^3}{ac} + \frac{c^3}{ab}.$$

This equation is then equivalent to:

$$2(a^2b^2 + b^2c^2 + a^2c^2) = a^4 + b^4 + c^4.$$

On the other hand, from $a + b + c = 0$, we get that $(a + b + c)^2 = 0$, or

$$a^2 + b^2 + c^2 = -2(ab + bc + ac).$$

Squaring again yields:

$$a^4 + b^4 + c^4 + 2(a^2b^2 + b^2c^2 + a^2c^2) = 4(a^2b^2 + b^2c^2 + a^2c^2) + 8abc(a + b + c),$$

or

$$a^4 + b^4 + c^4 = 2(a^2b^2 + b^2c^2 + a^2c^2),$$

as required.

15. Let $m \geq 100$ be an integer. Flavia writes the numbers $m, m + 1, m + 2, \ldots, 2m$ each on different cards. She then shuffles these $m + 1$ cards and divides them into two piles. Prove that at least one of the piles contains two cards such that the sum of their numbers is a perfect square.

Solution

It is enough to find three cards numbered x, y, and z such that $x+y$, $x+z$, and $y+z$ are perfect squares. This is because two of the numbers x, y, or z will be in the same pile. Solving the system of equations:

$$x+y=(2t-1)^2, \quad x+z=(2t)^2, \quad y+z=(2t+1)^2,$$

yields

$$x=2t^2-4t, \quad y=2t^2+1, \quad z=2t^2+4t.$$

Using these values of x, y, and z it suffices to show that for each integer $m \geq 100$, there exists a positive integer t such that $m \leq 2t^2-4t$ and $2t^2+4t \leq 2m$. That is,

$$t^2+2t \leq m \leq 2t^2-4t.$$

To this end, let t be the greatest integer such that $t^2+2t \leq m$. Note that $t \geq 9$ because $m \geq 100$. Since t is the greatest integer satisfying $t^2+2t \leq n$, it follows that

$$(t+1)^2+2(t+1) \geq m+1 \quad \text{if and only if} \quad m \leq t^2+4t+4.$$

To complete the proof, it is enough to show that $t^2+4t+4 \leq 2t^2-4t$. This is equivalent to $(t-4)^2 \geq 20$. But this is true since $k \geq 9$.

Some important remarks:

- The above proof shows that the conclusion of the problem is true for all $m \geq 99$.
- The conclusion of the problem is false for $m=98$. Indeed, a counterexample occurs if the first pile contains the even numbers from 98 to 126, the odd numbers from 129 to 161, and the even numbers from 162 to 196, while the second pile contains the rest of the numbers.